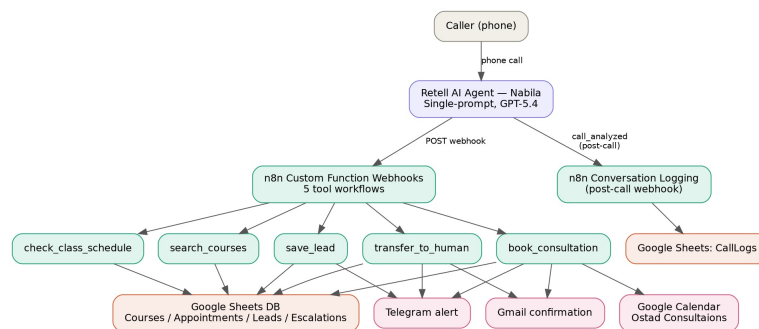


Ostad Academy AI Voice Receptionist

Ostad Academy - AI Voice Receptionist

An AI-powered voice receptionist for Ostad Academy, built with **Retell AI** (voice agent), **n8n** (automation backend), and **Google Sheets** (live database). The agent answers real phone calls, searches courses, checks class schedules, books consultations, captures leads, and escalates to a human when needed - entirely by voice, in English or Bangla.

Architecture



Architecture diagram

Flow summary:

1. A caller dials the Retell phone number and talks to **Nabila**, the voice agent.
2. Nabila calls one of 5 **Custom Function** tools over HTTP whenever she needs real data or needs to take an action - she never invents course names, fees, or dates.
3. Each tool is a webhook into **n8n**, which runs a dedicated workflow, reads/writes **Google Sheets**, and returns a spoken-friendly summary back to Retell.
4. After every call ends, Retell fires a separate **post-call webhook** with the full transcript and Retell's own sentiment/intent analysis, which n8n logs to the **CallLogs** tab automatically.

Tools (Retell Custom Functions)

Function	Purpose	Key inputs
search_courses	Look up course name, fee, duration, batch date	query
check_class_schedule	Get next batch start date and class days	course_name
	Book a free	name, phone,

book_consultation	consultation, checks slot conflicts	preferred_date, preferred_time
save_lead	Capture an interested caller who didn't book	name, phone, email, interested_course
transfer_to_human	Escalate to staff on complaint or explicit request	caller_name, phone, reason, urgency

All 5 are implemented as **n8n webhook nodes** in one workflow (*Ostad AI Receptionist - Custom Function Webhooks v2*), each calling a dedicated **sub-workflow** that owns the actual business logic and Google Sheets read/write. Every response is returned as { "spoken_summary": "...", "success": true|false } so the agent's prompt can read a consistent, predictable field name.

Bonus features implemented (3 of 8)

Feature	How it works
Email confirmation	Every successful booking sends a confirmation email via Gmail with caller name, phone, date, and time.
Google Calendar integration	Every successful booking creates a real event on the Ostad Consultations calendar, 1-hour slot, correct timezone.
Telegram notifications	Real-time Telegram alerts fire on 3 key events: new booking, new lead, and human handoff - so staff never miss a live signal.

Conversation logging

A separate n8n workflow listens on Retell's **Agent Webhook URL** for the call_analyzed event (fires once per call, after Retell's own analysis completes) and logs to the **CallLogs** tab:

- Call ID, caller number, timestamp, duration
- Intent (from a custom post-call analysis field)
- Sentiment (from Retell's built-in analysis)
- Full transcript and summary

Data store - Google Sheets

One spreadsheet, 5 tabs, acting as the live database:

Tab	Columns
Courses	CourseName, Category, Duration, Fee, BatchStart, Schedule, Instructor, SeatsLeft
Appointments	AppointmentID, Timestamp, Name, Phone, Email, Course, Date, Time, Status, CalendarEventID
	LeadID, Timestamp, Name,

Leads	Phone, Email, InterestedCourse, Source
Escalations	Name, Phone, Reason, Status
CallLogs	CallID, Timestamp, CallerNumber, DurationSeconds, Duration, Intent, Sentiment, Summary, Transcript

Error handling

- Every n8n tool workflow has an explicit error branch - if Google Sheets, Calendar, or Gmail fails, the caller still gets a graceful spoken fallback (“let me have an advisor confirm that”) instead of a dead call.
 - Booking conflicts are checked before writing - a taken slot returns alternative-time guidance instead of a false confirmation.
 - Calendar and email steps use `continueRegularOutput` on error, so a failed side-integration never blocks the core booking confirmation.
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Prompt design highlights

- Tool-first instruction: the agent is told to call a tool immediately on any course question rather than asking clarifying questions first.
 - Strict escalation criteria: human handoff only triggers on explicit request, real complaints, or genuine frustration - not on ordinary uncertainty.
 - Bilingual: mirrors English/Bangla mid-call without announcing the switch.
 - Guardrails against inventing fees, dates, or seat counts not returned by a tool.
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Tech stack

- **Voice agent:** Retell AI (single-prompt agent, GPT-5.4)
- **Automation backend:** n8n (Cloud)
- **Database:** Google Sheets
- **Integrations:** Google Calendar, Gmail, Telegram Bot API